

CSCI 112 Assignment – Standard Library Functions II

Points: 10

Automated tests will run on GitHub Actions after every push. Check the Actions tab to see which tests pass and which fail.

Instructions

1. **Fork** the repository on GitHub - <https://github.com/jpach-cs/112-lab04-template>.
2. Clone your forked repository to your local machine.
3. Open the file `code.c`.
4. Implement every function listed below:
 - Read the comments above each function carefully.
 - Follow the rules — use **ONLY** `if`, `switch`, `goto`, `putchar()`, `getchar()`, `sprintf()`.
 - Do NOT use `while`, `for`, `do-while`, or any standard library functions.
 - `my_isalpha()` MUST call your own `my_islower()` and `my_isupper()`.
 - `my_isalnum()` MUST use `switch()` and call your own `my_isalpha()` and `my_isdigit()`.
 - Every format function MUST call `clear_string(dest, 64)` as its first statement.
5. Test your implementation locally using `main.c`.
6. Once all functions are complete — save your changes, then **commit** and **push** to your repository.
7. Go to the **Actions** tab on GitHub and check the test results.

Functions to implement

Utility

Function	Description
<code>void clear_string(char s[], int n)</code>	Sets first <code>n</code> characters of <code>s</code> to <code>'\0'</code> . MUST use <code>goto</code> .

From previous lab — re-implement

Function	Description
<code>int my_isdigit(char c)</code>	Returns 1 if <code>c</code> is '0'-'9', otherwise 0.
<code>int my_islower(char c)</code>	Returns 1 if <code>c</code> is 'a'-'z', otherwise 0.

New functions

Function	Description
<code>int my_isupper(char c)</code>	Returns 1 if <code>c</code> is 'A'-'Z', otherwise 0.
<code>int my_isalpha(char c)</code>	Returns 1 if <code>c</code> is a letter. MUST call <code>my_islower()</code> and <code>my_isupper()</code> .
<code>int my_isalnum(char c)</code>	Returns 1 if <code>c</code> is a letter or digit. MUST use <code>switch()</code> .
<code>int my_strcmp(char a[], char b[])</code>	Returns -1, 0, or 1.
<code>int my_strchr(char s[], char c)</code>	Returns index of first occurrence of <code>c</code> , or -1.
<code>int my_pow(int a, int b)</code>	Returns <code>a</code> raised to the power of <code>b</code> .
<code>double my_pow_double(double a, int b)</code>	Returns <code>a</code> raised to the power of <code>b</code> .

Format functions

Function	Description
<code>char * format_my_isupper(char dest[], char c, int r)</code>	e.g. "isupper('A') = true"
<code>char * format_my_isalpha(char dest[], char c, int r)</code>	e.g. "isalpha('a') = true"
<code>char * format_my_isalnum(char dest[], char c, int r)</code>	e.g. "isalnum('!') = false"
<code>char * format_my_strcmp(char dest[], int r)</code>	e.g. "comparison: equal"
<code>char * format_my_strchr(char dest[], int r)</code>	e.g. "found at: 2" or "not found"
<code>char * format_my_pow(char dest[], int r)</code>	e.g. "pow = 256" — left-justified, width 12
<code>char * format_my_pow_double(char dest[], double r)</code>	fixed total width of 12, adaptive precision

Iteration pattern — reminder

```
int i = 0;
loop:
    if (i >= n)
        goto done;
    // body
    i++;
    goto loop;
done:
```

#Header comment — required in code.c

```
// Your Name
// CSCI 112 Fall 2026
// Programming Assignment #1
// I declare that I am the author of this work, take full responsibility for it, and have d
```
